

Satyendra Singh

Full Stack Developer | DevOps Engineer

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SUMMARY

Full Stack Developer and DevOps Engineer with 3+ years of experience designing, developing, and deploying scalable web and mobile applications using React.js, React Native, Node.js, Django, and modern JavaScript frameworks. Hands-on with DevOps tooling and practices including Docker, Kubernetes, AWS, Jenkins, Terraform CLI, Ansible, GitHub, and CI/CD pipelines for building, containerizing, and deploying end-to-end solutions across frontend, backend, APIs, and cloud infrastructure. Experienced in IoT and embedded systems integration, real-time communication, payment gateway integration, and enterprise-level access control systems. Proficient in delivering high-performance, reliable applications with a strong focus on automation, system reliability, and product innovation.

EDUCATION

Lovely Professional University	<i>Jalandhar, Punjab</i>
<i>Bachelor of Technology in Electronics (B.Tech)</i>	<i>Aug. 2013 – May 2018</i>
Aadarsh Sarvodaya Higher Secondary School	<i>Anuppur, M.P.</i>
<i>Science</i>	<i>Aug. 2011 – May 2012</i>
Aadarsh Sarvodaya High School	<i>Anuppur, M.P.</i>
	<i>Aug. 2009 – May 2010</i>

EXPERIENCE

Houston System	May 2023 – Present
<i>Software Engineer</i>	<i>Greater Noida, U.P.</i>
- Engineered a REST API using Node.js and MySQL to ingest and store data from 5+ learning management systems, processing over 10,000 records daily. - Architected a full-stack access control system using Node.js, React, MySQL, and Docker, integrating hardware like boom barriers, turnstiles, and tripods over TCP/IP, UDP, HTTP, WebSocket, and RS-485, deployed across 15+ entry points and 20+ concurrent devices with sub-200ms response times for 1,000+ daily users. - Integrated WebSocket, GraphQL, Apollo Client and Server, and Apache Kafka for real-time communication and event-driven architecture, reducing data sync latency by approximately 40%. - Built a complete CI/CD pipeline using GitHub, Jenkins, Docker, and Kubernetes to deploy on AWS infrastructure provisioned with Terraform CLI and Ansible, cutting manual deployment effort by approximately 60%, with Prometheus and Grafana for CPU and system health monitoring.	

PROJECTS

Intelligent Toll Management System <i>React.js, Redux Toolkit, Node.js, MySQL, ANPR</i>	Nov 2025 - Apr 2026
- Architected a large-scale intelligent toll management system for an 8-lane toll plaza, consisting of 4 upward-direction lanes and 4 downward-direction lanes, enabling seamless vehicle movement and streamlined toll collection. - Crafted responsive and real-time frontend modules using React.js and Redux Toolkit for centralized lane monitoring, transaction tracking, and operator management. - Incorporated Airtel Payments Bank FASTag services to process vehicle toll transactions in real time by securely sending FASTag data to banking APIs for validation, balance deduction, and transaction authorization. - Designed automatic gate control logic where boom barriers open instantly upon successful FASTag verification and payment confirmation, significantly reducing vehicle waiting time. - Unified multiple hardware systems including ANPR cameras, AVCC systems, lane displays, RFID readers, handheld ticketing devices, and boom barriers to enable synchronized toll operations and real-time monitoring. - Created RESTful backend services using Node.js and MySQL to manage lane data, transaction records, vehicle logs, operator activities, and hardware communication across the toll infrastructure. - Enhanced operational efficiency by implementing real-time device communication, centralized monitoring dashboards, and streamlined vehicle identification workflows for high-volume traffic handling.	

Smart Control Mobile Application | *React Native, Node.js, MQTT, WiFi, Bluetooth* June 2025 – October 2025

- Engineered a cross-platform mobile application using React Native for real-time monitoring and control of 50+ connected IoT devices over WiFi.
- Enabled MQTT-based communication between the mobile application and ESP32-based controllers, achieving device response times under 200ms for low-latency control.
- Planned room-wise device management across 10+ rooms with live status synchronization and remote access functionality.
- Combined secure authentication and cloud connectivity using Firebase and REST APIs, supporting 100+ registered users for remote access and user management.
- Delivered a scalable IoT architecture supporting 50+ connected devices with real-time communication and under 1% message loss.

Smart Parking Management System | *React.js, Node.js, MySQL, Redis, Kafka, AWS* Sep 2024 - March 2025

- Developed a full-stack parking management system supporting 500+ subscribed vehicles, using React.js and Tailwind CSS, and integrating boom barriers and P10 LED displays with the Node.js/MySQL backend over TCP/IP on an HTTPS/SSL-secured server.
- Designed subscription-aware FASTag billing logic—active subscribers were exempted from deductions while non-subscribers were auto-charged—via Airtel Payments Bank FASTag integration and real-time payment webhooks.
- Leveraged Apache Kafka and Redis for event-driven, low-latency processing, with sockets enabling real-time bi-directional communication between hardware devices and the backend for instant barrier control.
- Automated reconciliation of pending FASTag transactions via a daily scheduled job that queries the bank API for deduction status, ensuring accurate subscription-based billing.
- Containerized and deployed the application through a Jenkins-driven CI/CD pipeline on Kubernetes, with AWS infrastructure provisioned via Terraform CLI and Ansible for automated, zero-downtime releases.

Mullanpur Stadium Access Control System | *Node, MySQL, Indexing, T20 machine* Feb 2024 - Aug 2024

- Planned and deployed an access control system across multiple gates in the Mullanpur Cricket Stadium for various categories including BCCI officials, Punjab Kings team members, service staff, and ticket holders.
- Implemented an efficient ticket validation process by pairing QR and RFID scanners at entry points, leading to the processing of over 10,000 tickets per event with a 98% accuracy rate in validation.
- Defined gate-level authorization logic—e.g., team gates exclusively allow Punjab Kings team cards; BCCI gates only allow BCCI officials.
- Ensured real-time validation and access control by connecting scanning systems to a centralized Flask-based backend and MySQL database.
- Optimized gate operations by ensuring that users are only allowed through gates based on their access type and ticket permissions.

Houston System Company Website | *React.js, Tailwind CSS, TypeScript, cPanel, Git* June 2023 – January 2024

- Delivered a responsive and SEO-friendly company website for Houston System to showcase products, solutions, and services.
- Implemented the frontend using React.js, Tailwind CSS, and TypeScript for a clean, modern, and mobile-friendly UI.
- Created modular and reusable UI components to ensure consistency and ease of maintenance across the site.
- Connected backend services using Node.js and Express to manage contact forms, product data, content updates, and a chatbot for real-time customer interaction. Rolled out WhatsApp chat integration to collect customer feedback directly through the website.
- Configured version control and code collaboration workflows via Git, ensuring streamlined development and peer reviews.
- Purchased and configured the company domain and deployed the production website via cPanel hosting, managing DNS records, SSL, and file deployment for a live public-facing site.

TECHNICAL SKILLS

Languages: Python, C/C++, MySQL, JavaScript, TypeScript, Flutter, HTML/CSS

Frameworks: React.js, Node.js, React Native, Angular, Flask, Django, Material-UI, Bootstrap, Tailwind CSS

DevOps & Cloud: GitHub, Docker, Kubernetes, AWS (EC2, S3, IAM), Jenkins, Terraform CLI, Ansible, CI/CD, Prometheus, Grafana

Messaging & Real-Time: Apache Kafka, Redis, Socket.io, WebSocket, Webhooks, GraphQL, MQTT

Developer Tools: Git, VS Code, Visual Studio, PyCharm

Libraries: pandas, NumPy, Matplotlib